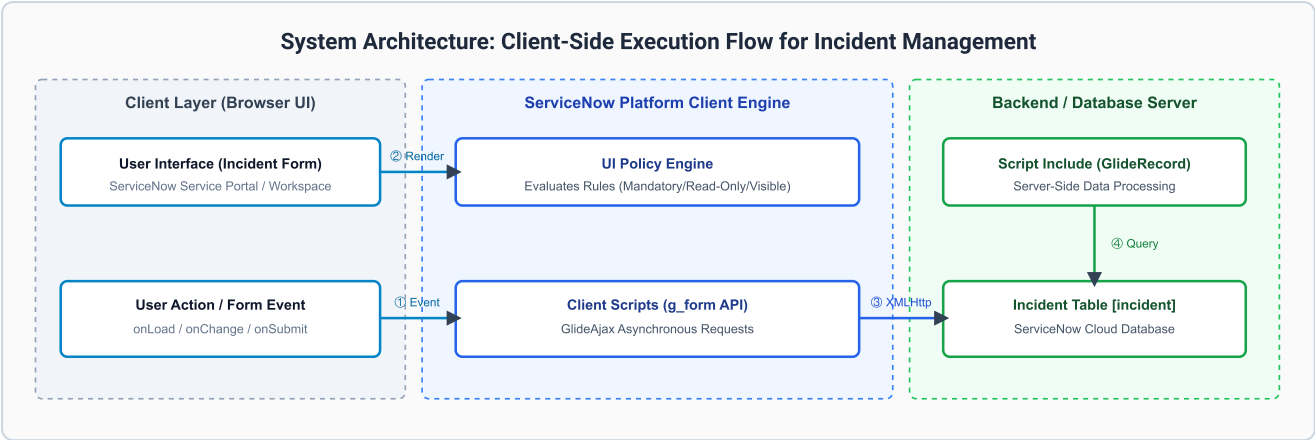


Date	30 September 2026
Team ID	6ab9453830f9767f8773c674
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture

The deliverable details the end-to-end client-side scripting and UI policy execution architecture on the ServiceNow ITSM platform. The execution flow spans client browser interaction, Client Script event handlers, UI Policy engine evaluations, GlideAjax asynchronous server calls, and ServiceNow cloud data storage.



Guidelines:

- Includes all client-side form event processes (onLoad, onChange, onSubmit, onCellEdit).
- Provides clear infrastructural demarcation between Client Browser and ServiceNow Cloud Platform.
- Indicates external/internal API interfaces (GlideAjax, REST/SOAP APIs for third-party integrations).
- Indicates data storage components (Incident Table `incident`, User Table `sys\_user`).

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the fulfiller/user interacts with the Incident form (Service Portal, Next Experience Workspace, Classic UI).	HTML5, CSS3, JavaScript (ES6+), AngularJS, ServiceNow Jelly / UI Macros
2.	Client Scripting Logic	Client-side script execution handling events: `onLoad`, `onChange`, `onSubmit`, and `onCellEdit`.	ServiceNow `g_form` API, `g_user` API, Client Scripts (JavaScript)
3.	UI Policy Engine	Declarative condition evaluation engine to dynamically control field visibility, mandatory status, and read-only attributes.	ServiceNow UI Policies & UI Policy Actions (Zero-code / Client Execution)

S.No	Component	Description	Technology
4.	Asynchronous Server Logic	Server-side helper script executed asynchronously from client scripts to fetch caller, CI, or asset details.	ServiceNow `GlideAjax`, Server-Side `Script Include` (JavaScript)
5.	Database / Storage	Core transactional data tables storing Incident records, Audit logs, and System dictionary configurations.	ServiceNow Cloud Database (MariaDB/MySQL relational backend)
6.	Cloud Storage	Storage for incident attachments, logs, dynamic form assets, and systemic update sets.	ServiceNow Cloud Infrastructure Storage / Advanced File Storage
7.	External API Interfaces	Integrations to pull third-party monitoring alerts or push incident resolution webhooks to external enterprise systems.	REST API, SOAP Web Services, JSON / XML Payload Parsing
8.	Automation & AI Intelligence	Automated incident priority calculation based on Impact/Urgency matrix and AI predictive intelligence for assignment.	ServiceNow Business Rules, Data Lookup Definitions, Predictive Intelligence ML
9.	Infrastructure & Server	Multi-instance cloud architecture providing high-availability ServiceNow instance hosting.	ServiceNow Enterprise Cloud Platform (SaaS / PaaS), Advanced High Availability (AHA)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source & Platform Frameworks	Client-side libraries and platform abstraction frameworks utilized for form rendering and validation scripts.	JavaScript (ES6), Prototype.js, AngularJS, Bootstrap (Portal UI)
2.	Security Implementations	Role-Based Access Control (RBAC), Contextual Security, field-level ACLs enforced alongside client UI policies to prevent unauthorized data tampering.	Access Control Lists (ACLs), TLS 1.3 Encryption, IAM, OAuth 2.0, OWASP Standards
3.	Scalable Architecture	Asynchronous `GlideAjax` client execution prevents UI blocking. Multi-tier cloud architecture scales to support thousands of concurrent IT agents.	Asynchronous JS Execution, ServiceNow Multi-Instance Cloud Architecture
4.	Availability	High-availability cloud environment with automated failover and zero-downtime upgrades across primary and secondary data centers.	Advanced High Availability (AHA), Load Balancing, Multi-Region Redundancy
5.	Performance Design	Optimized client scripts (avoiding synchronous `g_form.getReference`), client-side caching of UI policies, and CDN edge content delivery.	GlideAjax, Edge CDN Caching, Client-Side Rule Caching Engine

References:

- <https://developer.servicenow.com/> (Client Scripts & UI Policies Documentation)
- <https://c4model.com/> (Software Architecture Diagramming)
- <https://docs.servicenow.com/> (ServiceNow Product Documentation)
- <https://www.ibm.com/cloud/architecture> (Enterprise Cloud System Architecture)

